Direct Lyapunov MPC With Frozen Output Disturbance

Internal Technical Note

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1 Motivation and Scope

The previous frozen-`dhat` experiment path coupled three ideas in one workflow:

1. an offset-free upstream MPC candidate,
2. a first-step Lyapunov acceptance test, and
3. a replacement controller triggered only when the candidate failed.

That architecture was useful as a transition study, but it mixed two control laws in a single rollout and made the mathematical interpretation awkward. The rewrite in this note retires that transitional path and replaces it with a single method:

1. freeze the output disturbance estimate,
2. compute a steady target from that frozen disturbance, and
3. solve one direct Lyapunov MPC problem at every time step.

The new baseline is therefore easier to describe, easier to instrument, and closer to the final controller structure that should be compared in future studies.

2 Model and Coordinate Conventions

The controller-side model is used in scaled deviation coordinates with output disturbance only:

$$x_{k+1} = Ax_k + Bu_k, \tag{1}$$
$$y_k = Cx_k + d_k, \tag{2}$$
$$d_{k+1} = d_k. \tag{3}$$

Here:

- x_k is the physical controller state in deviation coordinates,
- u_k is the manipulated input in deviation coordinates,
- d_k is an output-offset disturbance state,
- y_k is the controlled output,

- the observer estimate is stored as the augmented vector

$$\hat{z}_k = \begin{bmatrix} \hat{x}_k \\ \hat{d}_k \end{bmatrix}.$$

The rewrite intentionally excludes any $A_{xd} d_k$ term from the state dynamics. In the code this is validated explicitly before a frozen-output-disturbance target is accepted.

3 Frozen Output-Disturbance Steady-State Target

3.1 Definition

At each time step k, the disturbance estimate is frozen:

$$d_s = \hat{d}_k. \tag{4}$$

Only x_s and u_s are solved online. The steady-state equations are

$$(I - A)x_s - Bu_s = 0, \tag{5}$$

$$Cx_s + d_s = y_{sp,k}. \tag{6}$$

After substituting $d_s = \hat{d}_k$, the output equation becomes

$$Cx_s = y_{sp,k} - \hat{d}_k. \tag{7}$$

3.2 Stacked Exact Form

The unbounded target solve is written as

$$\begin{bmatrix} I - A & -B \\ C & 0 \end{bmatrix} \begin{bmatrix} x_s \\ u_s \end{bmatrix} = \begin{bmatrix} 0 \\ y_{sp,k} - \hat{d}_k \end{bmatrix}. \tag{8}$$

The implementation supports:

- direct exact solves when the stacked matrix is square and numerically acceptable,
- least-squares fallback when the system is rectangular or rank-deficient,
- explicit diagnostics for rank, singular values, residual norms, and conditioning.

3.3 Reduced Form

When $I - A$ is invertible,

$$x_s = (I - A)^{-1} B u_s, \tag{9}$$

which induces the reduced output map

$$G u_s = y_{sp,k} - \hat{d}_k, \qquad G := C(I - A)^{-1} B. \tag{10}$$

This reduced representation is used for diagnostics and can also guide bounded least-squares solves when it is numerically attractive.

3.4 Bounded Variant

For bounded targets the same physics are preserved, but the steady input is constrained:

$$u_{\min} \le u_s \le u_{\max}. \tag{11}$$

The clean version deliberately avoids artificial anchors such as

$$\|x_s - \hat{x}_k\|^2, \qquad \|x_s - x_{s,\text{prev}}\|^2, \qquad \|u_s - u_{s,\text{prev}}\|^2.$$

Instead, the bounded solve minimizes only dynamic and output residuals:

$$\min_{x_s,u_s} \|(I-A)x_s - Bu_s\|^2_{Q_{\text{dyn}}} + \left\|Cx_s - (y_{sp,k} - \hat{d}_k)\right\|^2_{Q_y} \tag{12}$$

subject to the input box constraints above.

This makes the bounded target interpretable: when the exact steady target is outside the admissible input box, the solver finds the closest admissible steady-state compromise and reports the remaining residual.

4 Direct Lyapunov MPC Formulation

4.1 Prediction Model

The direct controller freezes the disturbance within each online solve:

$$x_{i+1|k} = Ax_{i|k} + Bu_{i|k}, \tag{13}$$

$$y_{i|k} = Cx_{i|k} + \hat{d}_k, \tag{14}$$

with initial condition

$$x_{0|k} = \hat{x}_k. \tag{15}$$

The steady target (x_s, u_s, d_s) is produced first and then held fixed while the MPC optimization is solved.

4.2 Objective

The implementation uses a practical tracking objective:

$$J_k = \sum_{i=0}^{N_p-1} \|y_{i|k} - y_k^{\text{track}}\|^2_Q + \sum_{i=0}^{N_c-1} \|u_{i|k} - u_s\|^2_S + \sum_{i=0}^{N_c-1} \|\Delta u_{i|k}\|^2_R + \|x_{N_p|k} - x_s\|^2_{P_x}. \tag{16}$$

In the current notebook and rollout path, y_k^{track} defaults to the raw setpoint $y_{sp,k}$, while (x_s, u_s) define the steady target used by the terminal and Lyapunov terms.

4.3 Input and Terminal Constraints

The online problem enforces the standard input box constraints

$$u_{\min} \le u_{i|k} \le u_{\max}, \tag{17}$$

and it optionally enforces the terminal set condition

$$(x_{N_p|k} - x_s)^\top P_x (x_{N_p|k} - x_s) \le \alpha_k. \tag{18}$$

The scalar α_k is computed from the existing terminal-input admissibility logic already present in the Lyapunov core.

4.4 Lyapunov Function and First-Step Contraction

The physical-state tracking error is

$$e_{x,k} = \hat{x}_k - x_s, \tag{19}$$

and the Lyapunov function is

$$V_k = e_{x,k}^\top P_x e_{x,k}. \tag{20}$$

The direct controller enforces first-step contraction inside the MPC problem:

$$(x_{1|k} - x_s)^\top P_x (x_{1|k} - x_s) \le \rho V_k + \varepsilon_{\text{lyap}}. \tag{21}$$

This is the hard mode.

4.5 Soft Lyapunov Mode

To avoid falling back to the old candidate-and-replacement structure, the rewrite also provides a soft mode with one nonnegative slack:

$$(x_{1|k} - x_s)^\top P_x (x_{1|k} - x_s) \le \rho V_k + \varepsilon_{\text{lyap}} + \sigma_k, \tag{22}$$

$$\sigma_k \ge 0. \tag{23}$$

The slack is penalized directly in the objective:

$$J_k^{\text{soft}} = J_k + \lambda_\sigma \sigma_k, \tag{24}$$

where λ_σ is a large user-visible penalty. This preserves a single direct MPC solve even when the hard contraction inequality would be infeasible.

5 Algorithmic Procedure

At each control step k, the implementation executes:

1. read the current plant output and previous input,
2. update the observer residual and current augmented estimate,
3. freeze the disturbance estimate $d_s = \hat{d}_k$,
4. solve the steady target in either unbounded or bounded mode,
5. compute the terminal admissibility scalar α_k,
6. solve the direct Lyapunov MPC in hard or soft mode,
7. apply the first move,
8. propagate the observer and plant,
9. log target diagnostics, solver status, Lyapunov values, contraction margins, and slack usage.

The important difference from the retired notebook is that there is no upstream candidate controller whose solution is later replaced by another MPC solve. The single direct controller is the online policy.

6 Implementation Mapping

The current repository mapping is:

- `Lyapunov/frozen_output_disturbance_target.py`

  Provides the clean output-disturbance-only target APIs:

  - `solve_target_unbounded_output_disturbance(...)`,
  - `solve_target_bounded_output_disturbance(...)`,
  - `solve_output_disturbance_target(...)`.

- `Lyapunov/direct_lyapunov_mpc.py`

  Provides:

  - the direct MPC solver class built on top of the existing Lyapunov core,
  - hard and soft first-step contraction modes,
  - the closed-loop rollout function,
  - debug-bundle, summary-table, plotting, and artifact-export helpers.

- `DirectLyapunovMPC_FrozenOutputDisturbance.ipynb`

  This is the experiment entrypoint for configuring and running the method.

- `Simulation/run_mpc_first_step_contraction.py`

  The transitional frozen-target mode plumbing has been removed from this older runner so that the new direct path is the only frozen-output-disturbance implementation.

7 Diagnostics and Interpretation

The new workflow is designed to support two orthogonal comparisons:

1. **Unbounded vs bounded target mode.**

   This comparison isolates whether the steady target is naturally admissible or must be projected into the input box. The key diagnostics are residual norms, active bound indicators, and the fraction of steps that require the bounded fallback.

2. **Hard vs soft Lyapunov mode.**

   This comparison isolates feasibility pressure in the direct controller. The key diagnostics are the contraction margin, the fraction of steps satisfying the hard inequality, and the size and frequency of the soft slack.

   The direct bundle therefore records:

- steady targets (x_s, u_s, d_s, y_s),
- target rank and conditioning diagnostics,
- exact-vs-least-squares flags,
- Lyapunov values V_k, $V_{1|k}$, and the contraction bound,

- hard and relaxed contraction satisfaction flags,
- soft slack magnitude when applicable,
- stable step tables, summaries, and timestamped plot directories.

8 Conclusion

This rewrite establishes a cleaner baseline for the polymer CSTR Lyapunov experiments:

- frozen output-disturbance target generation with no extra target regularizers,
- a single direct Lyapunov MPC solve at every time step,
- explicit hard and soft contraction variants,
- notebook-level switching between bounded/unbounded targets and hard/soft Lyapunov modes,
- diagnostics and exports aligned with later comparison studies.

This should be the default path for future frozen-output-disturbance Lyapunov experiments and the reference implementation for subsequent tuning or analysis work.